INSTRUCTOR'S DISCUSSION GUIDE

Topic 8: Static Internet Protocol (IP) Configuration

Configure static IP addresses on Windows hosts, verify local vs. remote delivery, and test network reachability.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Total Discussion Units: 11 Slides

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Subnet Mask, Gateway, and DNS Roles

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The subnet mask defines the local network boundary, the default gateway forwards remote traffic to the outside world, and Domain Name System (DNS) resolves hostnames to numeric Internet Protocol (IP) addresses.

- 192.168.1.10/24 uses 24 network bits and 8 host bits; its mask is 255.255.255.0.
- The default gateway is a local router used for destinations without a more specific route.
- Domain Name System (DNS): maps names to records such as Internet Protocol (IP) addresses.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- 192.168.1.10/24 uses 24 network bits and 8 host bits; its mask is 255.255.255.0.
- The default gateway is a local router used for destinations without a more specific route.
- DNS: maps names to records such as IP addresses.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of subnet mask, gateway, and dns roles.

Key Takeaway: The subnet mask defines the local network boundary, the default gateway forwards remote traffic to the outside world, and Domain Name System (DNS) resolves hostnames to numeric Internet Protocol (IP) addresses.

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Two Ways to Assign an IP Address

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

An address can be configured manually or obtained through a configuration service. Static and dynamic assignment differ in how settings are supplied and maintained.

- Static: an administrator manually configures the address and related settings.
- Dynamic: Dynamic Host Configuration Protocol (DHCP) supplies an address lease and configuration options.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Static: an administrator manually configures the address and related settings.
- Dynamic: DHCP supplies an address lease and configuration options.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of two ways to assign an ip address.

Key Takeaway: An address can be configured manually or obtained through a configuration service. Static and dynamic assignment differ in how settings are supplied and maintained.

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What Is Static IP Configuration?

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Static configuration assigns an interface's address settings manually. A predictable address helps infrastructure and services remain locatable when settings are managed correctly.

- Record address, mask, gateway, and Domain Name System (DNS) before changing anything. A Dynamic Host Configuration Protocol (DHCP) reservation is a different way to provide a stable address.
- Choose a unique usable address in the intended subnet and coordinate it with the administrator's addressing plan.
- Avoid unmanaged overlap with Dynamic Host Configuration Protocol (DHCP) pools. Duplicate addresses cause conflicts even when each individual address is syntactically valid.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Record address, mask, gateway, and DNS before changing anything.
- A DHCP reservation is a different way to provide a stable address.
- Choose a unique usable address in the intended subnet and coordinate it with the administrator's addressing plan.
- Avoid unmanaged overlap with DHCP pools.
- Duplicate addresses cause conflicts even when each individual address is syntactically valid.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of what is static ip configuration?.

Key Takeaway: Static configuration assigns an interface's address settings manually. A predictable address helps infrastructure and services remain locatable when settings are managed correctly.

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Four Key IPv4 Configuration Settings

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A basic host configuration includes an address, subnet mask, default gateway, and name resolver. Each field has a separate purpose and must match the intended network.

- Internet Protocol (IP) address: a unique usable address assigned to the interface.
- Subnet mask: identifies which address bits describe the local network.
- Default gateway: a local router for destinations needing a remote route.
- Domain Name System (DNS) server: a resolver for names; not needed for numeric-IP local tests.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- IP address: a unique usable address assigned to the interface.
- Subnet mask: identifies which address bits describe the local network.
- Default gateway: a local router for destinations needing a remote route.
- DNS server: a resolver for names; not needed for numeric-IP local tests.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of four key ipv4 configuration settings.

Key Takeaway: A basic host configuration includes an address, subnet mask, default gateway, and name resolver. Each field has a separate purpose and must match the intended network.

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How Hosts Decide if a Peer is Local

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Each host uses its own address and mask to decide whether a peer is local. A successful local round trip needs compatible decisions and a working link in both directions.

- They can communicate by Internet Protocol (IP) on a correctly connected Local Area Network (LAN) without a default gateway.
- 192.168.1.10/24 and 192.168.2.20/24 need appropriate routing between the two subnets.
- Mismatched masks can make one host consider a destination local while the other sends through a gateway. Test both directions.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- They can communicate by IP on a correctly connected LAN without a default gateway.
- 192.168.1.10/24 and 192.168.2.20/24 need appropriate routing between the two subnets.
- Mismatched masks can make one host consider a destination local while the other sends through a gateway.
- Test both directions.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of how hosts decide if a peer is local.

Key Takeaway: Each host uses its own address and mask to decide whether a peer is local. A successful local round trip needs compatible decisions and a working link in both directions.

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Example: two manually configured classroom computers

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

Two hosts with unique usable addresses in the same subnet can communicate over a working local network. A default gateway is unnecessary for this numeric local exchange.

- Both hosts calculate 192.168.1.0/24 as their local network, so this exchange does not require a router.
- The addresses must be unique and the link and host policy must permit communication. A successful address calculation alone does not prove reachability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Configure static IP addresses on Windows hosts, verify local vs. remote delivery, and test network reachability.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Both hosts calculate 192.168.1.0/24 as their local network, so this exchange does not require a router.
- The addresses must be unique and the link and host policy must permit communication.
- A successful address calculation alone does not prove reachability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of example: two manually configured classroom computers.

Key Takeaway: Two hosts with unique usable addresses in the same subnet can communicate over a working local network. A default gateway is unnecessary for this numeric local exchange.

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When Subnet Masks Do Not Match

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Different subnet masks can make two hosts disagree about whether they share a local network. One host may try direct delivery while the other requires a route that is unavailable.

Interactive Demonstration: `asymmetricDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Calculate each host's local/remote decision independently using its own mask.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of when subnet masks do not match.

Key Takeaway: Different subnet masks can make two hosts disagree about whether they share a local network. One host may try direct delivery while the other requires a route that is unavailable.

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Configuring Static IP in Windows

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Windows adapter settings control an interface's address and related options. Configuration must be verified through both inspection and an appropriate connectivity test.

- Open Settings → Network & Internet → Advanced network settings → More network adapter options (or run ncpa.cpl).
- Open the intended adapter's Properties → Internet Protocol Version 4 (Transmission Control Protocol (TCP)/IPv4) → Properties. Enter the approved values, then confirm.
- Use ipconfig /all and ping the peer's numeric address. Record results; restore "Obtain an Internet Protocol (IP) address automatically" and automatic Domain Name System (DNS) if that was the original setup.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Open Settings → Network & Internet → Advanced network settings → More network adapter options (or run ncpa.cpl).
- Open the intended adapter's Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties.
- Enter the approved values, then confirm.
- Use ipconfig /all and ping the peer's numeric address.
- Record results; restore "Obtain an IP address automatically" and automatic DNS if that was the original setup.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of configuring static ip in windows.

Key Takeaway: Windows adapter settings control an interface's address and related options. Configuration must be verified through both inspection and an appropriate connectivity test.

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Windows Static Address Configuration Procedure

WALKTHROUGH

📖 WHAT TO READ & DISPLAY ON SCREEN

Manual address configuration requires the intended adapter and a coordinated set of valid settings. Recording the original configuration supports verification and restoration after the change.

- Record the original configuration and select the intended adapter. Open ncpa.cpl, adapter Properties, then Internet Protocol Version 4 (Transmission Control Protocol (TCP)/IPv4) Properties.
- On an approved isolated Local Area Network (LAN), 192.168.1.10/24 and 192.168.1.20/24 are compatible local hosts. A gateway is unnecessary for a numeric local test.
- Use ipconfig /all to inspect settings, then test the peer in both directions. If Internet Control Message Protocol (ICMP) is filtered, examine approved policy rather than disabling the firewall.
- Changing the second host to 192.168.2.20/24 requires appropriate gateways and bidirectional routing for cross-subnet communication.
- Restore the original approved Internet Protocol (IP)/Domain Name System (DNS) method after a physical demonstration; browser settings do not change the real Windows adapter.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Record the original configuration and select the intended adapter.
- Open ncpa.cpl, adapter Properties, then Internet Protocol Version 4 (TCP/IPv4) Properties.
- On an approved isolated LAN, 192.168.1.10/24 and 192.168.1.20/24 are compatible local hosts.
- A gateway is unnecessary for a numeric local test.
- Use ipconfig /all to inspect settings, then test the peer in both directions.
- If ICMP is filtered, examine approved policy rather than disabling the firewall.
- Changing the second host to 192.168.2.20/24 requires appropriate gateways and bidirectional routing for cross-subnet communication.
- Restore the original approved IP/DNS method after a physical demonstration; browser settings do not change the real Windows adapter.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of windows static address configuration procedure.

Key Takeaway: Manual address configuration requires the intended adapter and a coordinated set of valid settings. Recording the original configuration supports verification and restoration after the change.

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Static IP Compatibility and Testing

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Syntactically valid addresses do not guarantee that two hosts can communicate. Local subnet agreement, link availability, and any required routes determine the modeled path.

Interactive Demonstration: `static` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

Inspect host settings and assign a stable address to a shared network printer.

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Modeled network • validate fields, disconnect the link, or enable a router with interfaces in both subnets.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of static ip compatibility and testing.

Key Takeaway: Syntactically valid addresses do not guarantee that two hosts can communicate. Local subnet agreement, link availability, and any required routes determine the modeled path.

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Configuring a Static LAN Printer

APPLICATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A classroom host can use the same router address for its gateway and forwarded name-resolution service, while a printer benefits from an address coordinated outside the dynamic Dynamic Host Configuration Protocol (DHCP) pool.

- The host and gateway share a local subnet; the router may forward Domain Name System (DNS) requests.
- The printer benefits from an address excluded from the Dynamic Host Configuration Protocol (DHCP) dynamic pool to prevent conflicts.
- Verify numeric connectivity and service ports separately; a successful ping does not prove application reachability.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

☀️ REAL-WORLD ANALOGY / INTUITION

> *Inspect host settings and assign a stable address to a shared network printer.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- The host and gateway share a local subnet; the router may forward DNS requests.
- The printer benefits from an address excluded from the DHCP dynamic pool to prevent conflicts.
- Verify numeric connectivity and service ports separately; a successful ping does not prove application reachability.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of configuring a static lan printer.

Key Takeaway: A classroom host can use the same router address for its gateway and forwarded name-resolution service, while a printer benefits from an address coordinated outside the dynamic Dynamic Host Configuration Protocol (DHCP) pool.